LL3M: CCD-Gated Co-Design of a Ternary Language Model for One L3 Slice

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Abstract

Most "small" LMs still target phones or discrete GPUs with multi-hundred-megabyte payloads. We ask a different question: what if the product unit is one CPU last-level cache (L3) CCD? LL3M co-designs architecture, ternary/int8 quantization, portable C packing, curriculum, and host memory so the entire inference working set (weights + activations + KV) size-fits a single CCD gate (∼30.4 MiB on a 32 MiB slice).

The freeze is ∼79M parameters (L23×d576 MQA, T=256, V=2048), served as a 20.3 MiB L3T2 pack with continuous QAT matching C. Knowledge placement is first-class: ternary weights hold language and tool policy; facts live in a host Engram-lite KB and tools. On seven zero-shot MC tasks (n=400, denser-continue pack), macro is 41.4% vs 54.2% for Qwen3.5-0.8B-Base under portable weight-only int4 (−12.8 pp absolute; ∼9.5× more Qwen params; not iso-tokenizer/data). Tool-only SFT regresses SciQ by 18 pp; multi-mix SFT restores SciQ (macro 41.2%). On a closed 101-case agent suite (not BFCL), the tool-SFT ship pack scores 90/101 pure (assist and gate off; no_call 15/24) and 101/101 product only with assist off and a When2Call-style host gate on—product quality is model emission × host policy, not model-only competence. Artifact pins: checkpoints/paper_freeze_v072/ (SHA256 in MANIFEST.json).

On the reference host, C decode size-fits one CCD (runtime WS 24.3 MiB; design 30.2 MiB) with 100% mlock; hard+FIFO holds ∼280–282 tok/s. DRAM share of L2 fills is often WEAK without isolation; GOOD (≤5%) appears only under hard+FIFO / quiet-core (preferably isolcpus) paths—not a portable zero-DRAM guarantee. Multi-session packing under the same size gate reaches N=2 at full T (∼550 tok/s aggregate) and N=8 with short KV (∼2100–2200 aggregate). We name and operationalize CCD-gated co-design: a hard L3 size-fit constraint forcing joint shape/quant/serve/data/host choices, with train–serve lock, honest PMU, and dual product/pure agent metrics.

Keywords. CCD-gated co-design; L3-resident inference; ternary QAT; knowledge placement; tool-augmented SLMs; cache-first systems.

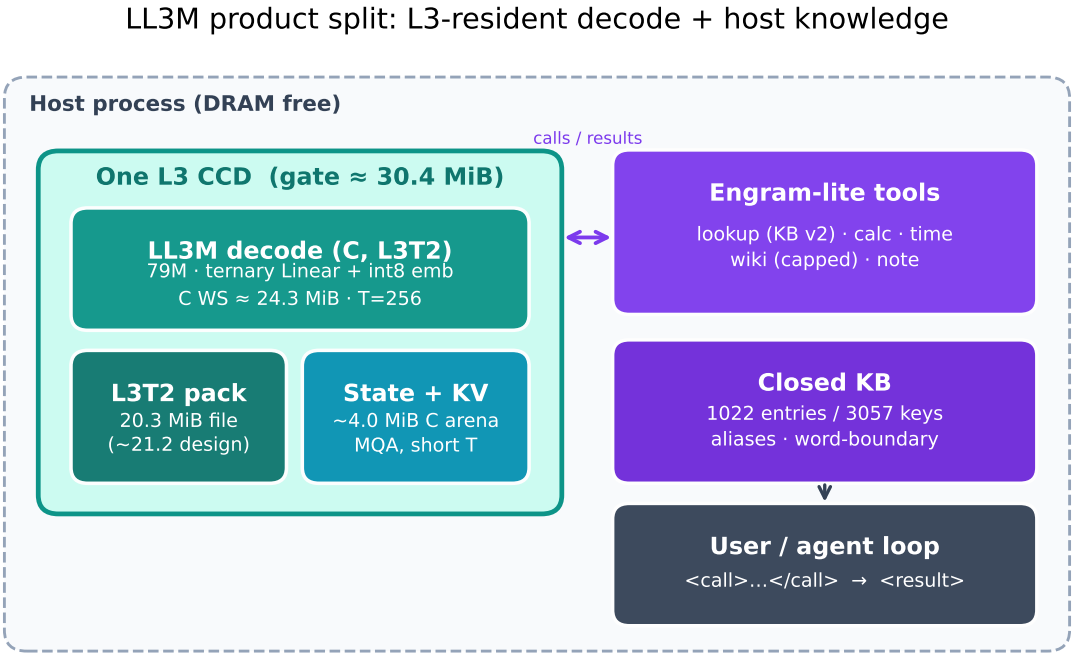


Figure 1: CCD-gated split. Decode stays inside one L3 CCD (L3T2 pack + state/KV); host Engram-lite tools own long-tail facts. C runtime WS ≈24.3 MiB; design estimate ≈30.2 MiB under a 30.4 MiB gate.

1 Introduction

Objective. Most small-LM work optimizes quality under a parameter or memory budget. We use a stricter unit: one CPU L3 CCD. If weights, activations, and KV size-fit that slice, residency hooks can keep the hot set off the DRAM path when the host cooperates—orthogonal to FLOP counts and to phone-GPU 4-bit footprints.

Why not just another tiny BitNet. Ternary nets [Ma et al., 2024, Wang et al., 2023], educational data [Gunasekar et al., 2023, Allal et al., 2025], on-device SLMs [Liu et al., 2024], and tool agents [Patil et al., 2023, Shim et al., 2025] each exist. Missing is an end-to-end product freeze that makes full-decode L3 residency non-negotiable: MQA at short T, tiny BPE, ternary+int8 packs, C residency hooks, host facts, and a dense multi-mix ladder under continuous QAT.

Thesis.

> Under a single-CCD inference gate, capability is maximized by co-designing shape, quant, runtime, data, and host memory—not by packing more facts into ternary weights.

Contributions.

1. CCD gate as objective: 0.95× one 32 MiB CCD (∼30.4 MiB); design WS ∼30.22 MiB; C runtime WS ∼24.3 MiB; on-disk pack 20.3 MiB for 78 815 712 ternary+int8 params (L23×d576 MQA, T=256, V=2048).
2. Train–serve quant lock: continuous ternary Linear + int8 embed QAT; portable C L3T2; residency hooks with measured size-fit and honest PMU (DRAM% host-dependent).

3. Knowledge placement: host Engram-lite KB (1022 entries / 3057 keys) + tools for facts; weights for language + tool policy.
4. Dense ladder + dual agent metrics: tool-only SFT harms SciQ (−18 pp); multi-mix restores SciQ; tool ship is 90/101 pure vs 101/101 product (host gate on) on a closed suite—not BFCL.

Claim → checkpoint map. Table 1 states which pack answers which metric (dual-SKU is intentional).

Table 1: Which checkpoint answers which claim (v0.7.9).

Claim	Checkpoint / pin	Metric
Closed-book macro 41.4%	v76_dense_c denser continue	7-task MC, n=400
SciQ peak 69.5%	v76_dense_c_p1 (C+)	SciQ / macro flat
Tool ship pure 90/101	pure_v2 + freeze pure pin	assist off, gate off
Tool ship product 101/101	pure_v2 + freeze product pin	assist off, gate on
KB snapshot	paper_freeze_v072/kb_facts.json	1022 / 3057
Size-fit / tok/s	denser-C L3T2 on ref. host	WS 24.3 MiB; ∼280 tok/s

Non-goals. We do not claim MMLU parity with 0.5B–8B models, L3-resident training, MoE, pure 1-bit weights, BFCL SOTA, or ∼0% DRAM share without isolation. We claim a measured operating point under a rare constraint.

2 Positioning

Table 2 states the contribution as a conjunction under a hard CCD gate, not any single brick.

Table 2: Novelty as constraint-driven conjunction (not isolated algorithms).

Axis	Typical SLM / BitNet	Typical tool agent	LL3M (this work)
Primary budget	params / VRAM / disk	API / latency	one L3 CCD WS
Serve claim	PyTorch / mobile NPU	cloud tools	portable C + residency
Quant story	post-hoc or train-only	often fp16 student	QAT locked to C pack
Knowledge	more pretrain tokens	RAG/tools add-on	placement under gate
Success metric	MMLU-style only	tool accuracy only	MC + agent + WS fit

Named principle. We name and operationalize CCD-gated co-design: a component is admissible only if the hot WS estimate and C pack respect the CCD gate; knowledge that cannot fit is externalized. This is a product constraint, not a new attention operator.

3 Related work

Data-efficient small LMs. Phi and SmolLM emphasize educational filtering and synthetic textbooks [Gunasekar et al., 2023, Li et al., 2023, Allal et al., 2025, Penedo et al., 2024, Eldan and Li, 2023]. MobileLLM studies sub-billion on-device shapes [Liu et al., 2024]. We keep the density intuition but subordinate it to a CCD gate, which forbids large vocabs and long contexts that those lines often retain.

Ternary / 1.58-bit training. BitNet b1.58 and bitnet.cpp [Ma et al., 2024, Wang et al., 2023, Microsoft Research, 2025, 2024] establish ternary STE and packing. LL3M's incremental systems contribution is native QAT through a multi-stage dense ladder with explicit checkpoint assertions and a C residency path under a fixed WS ceiling.

External memory and tools. Engram [DeepSeek-AI, 2026] argues for O(1) lookup beside neural compute. Tool-calling work [Patil et al., 2023, Shim et al., 2025, others, 2025, Chen and Li, 2024, others, 2026] shows small models need narrow schemas, gold SFT, and no-call negatives. We operationalize a lightweight Engram split for a 79M ternary model: host KB is mandatory product infrastructure, not an optional RAG demo.

Cache-aware inference. Kernel and paging systems [Gerganov and contributors, 2023, MLC team, 2023, Song et al., 2023] improve average-case locality. We instead use a hard product refusal: if estimated WS exceeds the CCD gate, the shape is rejected at design time; we further report measured mlock and PMU on the C path.

4 System design

4.1 Architecture

Table 3: LL3M product shape (v2 block).

Component	Choice
Block	RMSNorm, RoPE, SwiGLU, QK-Norm, no bias
Depth / width	23 layers, d=576, mid 1536
Attention	12 Q / 1 KV (MQA), d_h=48 [Shazeer, 2019, Ainslie et al., 2023]
Context / vocab	T=256, BPE V=2048
Stability	LayerScale 10^{-4}; peri-LN off
Train extras	z-loss [Chowdhery et al., 2022]; shared-head MTP [DeepSeek-AI, 2024] (train-only)
Parameters	78 815 712

Table 3 summarizes the freeze. At short T, MQA cheapens KV so depth/width absorb parameters under the gate. Peri-LN blocked early learning on this shape (val CE stuck ∼5.9 in short ablations); LayerScale alone progressed.

4.2 Quantization and pack

Linear layers use ternary STE (absmean) in train and 2-bit packing in C. Embeddings use int8 QAT: ternary embeds traded quality for a sub-MiB L3 win in earlier ablations. Export produces .ll3m (L3T2); denser-C on-disk pack is 20.32 MiB (C weights payload after load: 20.27 MiB). Training may use full DRAM/GPU; the product claim is C decode.

Post-hoc vs QAT (v75 lineage). Forcing ternary+int8 on an fp32 checkpoint raised val CE from 1.14 to 2.27; native ternary+int8 QAT on the same shape sits near val CE 1.26 on the serve grid. This supports continuous QAT over naive post-hoc pack.

Portable int4 baseline (Qwen). Per-row absmax fake-quant of Linear weights (not NF4, GPTQ, or NVFP4). For each row i, set $s_i = \max_j |w_{ij}|/7$. Round each scaled entry and clip to $q_{ij} \in \{-8, \dots, 7\}$, then set

$$\hat{w}_{ij} = s_i \, q_{ij}. \tag{1}$$

See Frantar et al. [2022], Dettmers et al. [2023] for related methods; we use a simpler portable weight-only protocol.

4.3 L3 residency

On the reference host (2×32 MiB L3 CCD), the product budget is $0.95 \times 32\,\text{MiB} \approx 30.4\,\text{MiB}$. Design-time estimate:

$$\text{WS}_{\text{design}} \approx W_{\text{weights}} + W_{\text{acts}} + W_{\text{KV}}. \tag{2}$$

Design estimator: ∼30.22 MiB (weights ∼21.2 + acts/KV ∼9.0). C runtime payload on denser-C: 24.32 MiB (weights 20.27 + state/KV arena 4.00 + pads). Soft/hard modes include CCD bind, mlock, hierarchical warm, optional isolcpus / FIFO / hugepages.

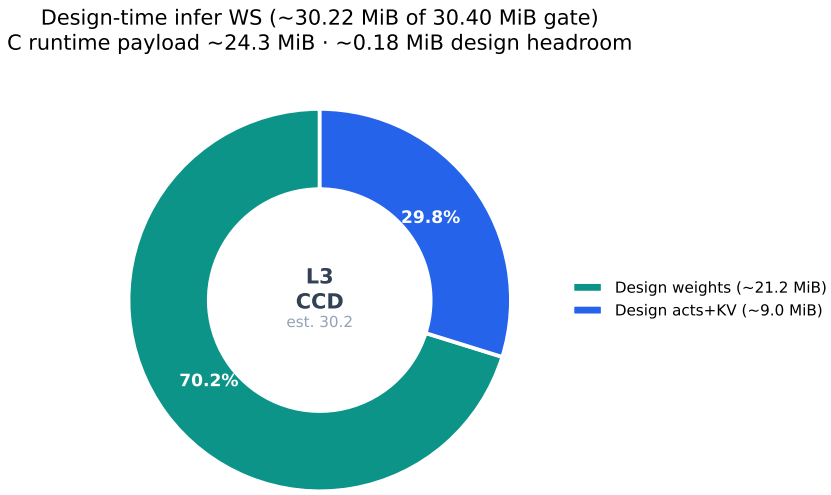


Figure 2: Design-time inference working-set estimate under the 30.4 MiB gate (shape freeze). C runtime measures a leaner ∼24.3 MiB payload on the same pack; both fit one 32 MiB CCD.

4.4 Host knowledge (Engram-lite)

Closed facts live in a JSON KB (v2: canonical ids, aliases, flat index; 1022 entries / 3057 keys). lookup uses word-bounded matching (avoid pi⊂pizza). Open-world queries use capped Wikipedia summaries; arithmetic uses host calc. The model emits tool-policy tokens; the host executes tools. Host table growth is free density relative to ternary weight bits. KB on/off (lookup subset, n=20, pure_v2 greedy): full KB → 20/20; empty KB → 0/20 under both product and pure flags (kb_onoff_ablation.json)—hits are host memory, not weight knowledge.

5 Training curriculum

Shape search. Exhaustive max-parameter search under the gate with practical head sets yields L23×d576 MQA.

Base English. We prefer educational web + dense synthetic cards over TinyStories-primary narrative [Eldan and Li, 2023]. Native ternary (+ int8 embed) QAT is kept on the product path.

Dense knowledge ladder.

1. A: Host KB expand/normalize.
2. B: ∼700M synthetic chars (atomic-biased) + ∼500M FineWeb-Edu (doc cap 1200); exact + MinHash near-dedup (Jaccard 0.85); ∼161M train tokens after BPE.
3. C: Continue ternary QAT +30k steps → denser continue.
4. C+: Further +15k dense continue under QAT.
5. D: Fact Q&A SFT then tool SFT (QAT on).
6. Mix: Multi-objective SFT from C: ∼50% dense / 20% fact / 30% tool, +5k steps, lr 3×10^{-5}.
7. Ship tools: tool-focused SFT lineage (pure_v2).

Every stage asserts checkpoint tags qat=ternary, embed_qat=int8. Dense ladder training primarily on remote CUDA; C residency on the local Ryzen L3 reference machine.

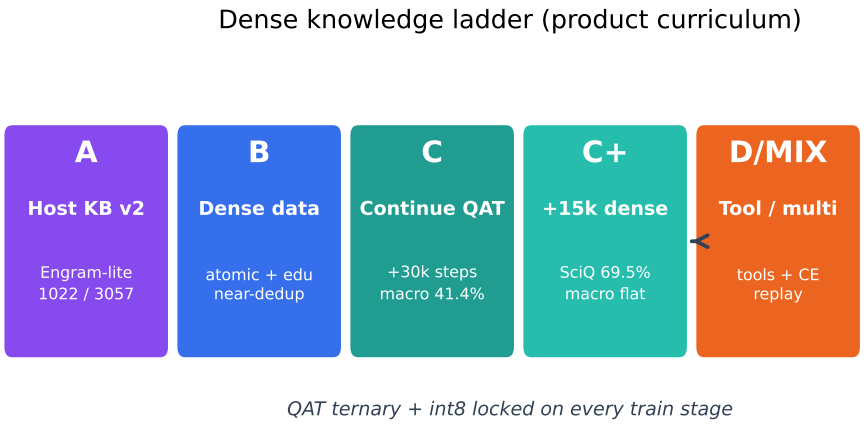


Figure 3: Dense knowledge ladder: host KB → atomic/edu data → ternary continue → C+ → tool/multi-mix SFT (QAT locked throughout).

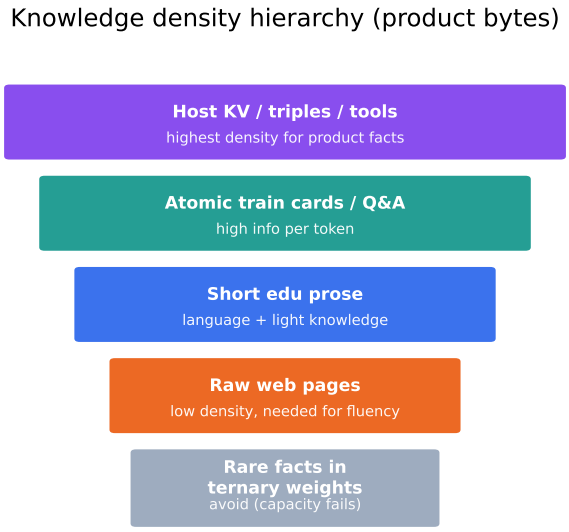


Figure 4: Knowledge density hierarchy for this product: prefer host tables and atomic cards over rare facts in ternary weights.

6 Evaluation

MC suite. Zero-shot continuation log-likelihood ranking, seed 42, limit 400 examples (COPA uses the full set n=100): HellaSwag, ARC-Easy, WinoGrande, BoolQ, OpenBookQA, SciQ, COPA. Macro accuracy is the unweighted mean of task accuracies. Domain BPE V=2048 vs Qwen's large vocab makes this a harsh, not iso-tokenizer, comparison. Primary tables use seed 42, limit 400. Bounded multi-seed probe (SciQ only, limit 100, seeds 42/43/44 on denser-C): acc 62/60/60% (mean 60.7 ± 1.2); paper SciQ 66.5% is limit 400 seed 42—not interchangeable.

Baseline. Qwen3.5 Base family under portable weight-only int4 (equation above), same tasks, limits, and seed. Primary reference: 0.8B-Base; size ladder also reports 2B/4B/9B. These are strong industry bases, not L3-resident packs. Canonical compare JSON lives in the repository compare tree (Appendix B).

Agent battery. Closed suite (101 cases): no-call chat, calc, lookup, time, language openers, adversarial, tokenizer check. Not BFCL / public When2Call leaderboard. We always report two flags: pure (assist off, gate off; model-only stress) and product (assist off, gate on; ship = model emission × host gate). Gold-inject assist is ablation only. Decode is C L3T2 unless noted. Primary ship pack is pure_v2; denser-C pure (42/101) is a language baseline without tool policy. Pins: Appendix B / paper_freeze_v072/MANIFEST.json. Product no-call (24/24) vs pure freeze no-call (15/24) is the main pure↔product gap on the ship pin. Gate instrumentation (§7.3) logs false-call blocks and tool rewrites; product scores remain system scores.

Residency card. C binary with --l3-resident / --residency-hard, hierarchical warm, CCD bind, optional PMU window after steady-warm. Logs under checkpoints/compare/.

7 Results

7.1 Closed-book macro

Table 4: Macro accuracy (7 tasks, n=400; COPA n=100). LL3M packs are 20.3 MiB L3T2. Qwen rows use portable weight-only int4.

Model	Params (approx.)	Macro
Qwen3.5-0.8B-Base int4	∼0.75B	54.21%
Qwen3.5-2B-Base int4	∼1.9B	59.39%
Qwen3.5-4B-Base int4	∼4.2B	64.96%
Qwen3.5-9B-Base int4	∼9.0B	69.11%
LL3M prior tool path	∼79M tern.	40.11%
LL3M denser continue (C)	∼79M tern.	41.39%
LL3M dense continue +15k (C+)	∼79M tern.	41.36%
LL3M tool/fact SFT (D)	∼79M tern.	40.46%
LL3M multi-mix SFT (MIX)	∼79M tern.	41.18%

Table 4 and Figure 5 summarize macros. C vs Qwen 0.8B int4: −12.82 pp absolute on this suite (∼9.5× Qwen params; portable absmax int4, not NF4/GPTQ; not iso-tokenizer). Relative deficit $1 - 41.39/54.21 \approx 23.6\%$ is secondary and easy to over-read as near-parity. Vs 9B int4 the absolute gap is −27.7 pp. C+ does not lift macro despite SciQ gains (Table 5). LL3M remains an L3-class pack (20.3 MiB) under a CCD gate that the Qwen bases do not target (Figure 6).

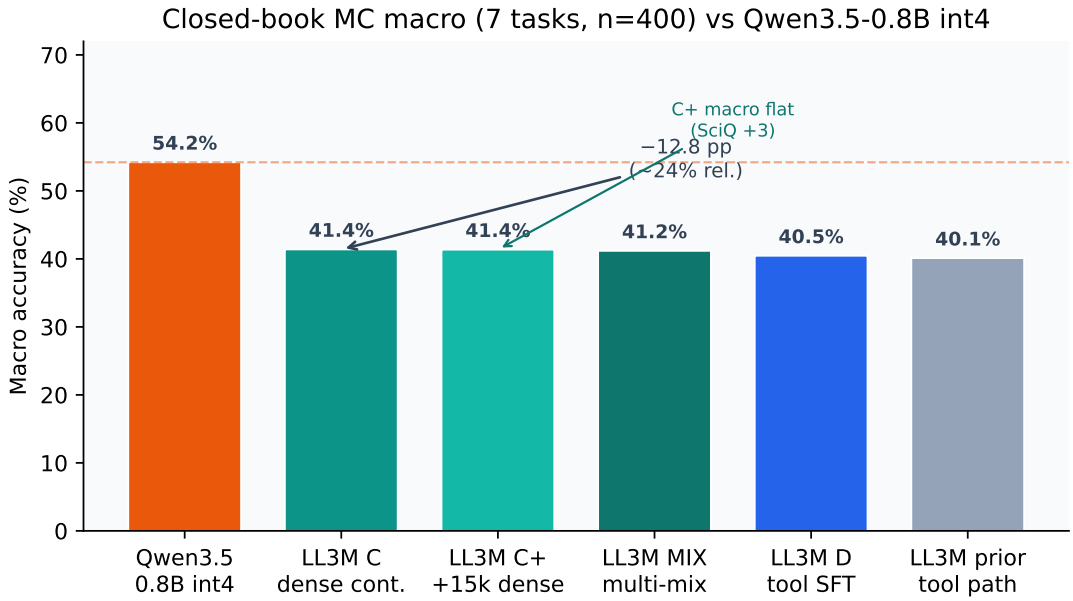


Figure 5: Macro accuracy on seven standard MC tasks (n=400). LL3M denser continue (C) reaches 41.4% versus 54.2% for Qwen3.5-0.8B-Base weight-only int4.

7.2 Per-task detail

Table 5: Per-task accuracy (%). Qwen column is portable weight-only int4 (macro 54.21%). Random ≈25–50% depending on the task. COPA n=100.

Task	C	C+	D	MIX	Qwen 0.8B int4	Rand.
HellaSwag	28.0	28.5	26.5	29.0	35.5	25
ARC-Easy	27.3	29.8	31.0	27.8	59.0	25
WinoGrande	47.5	48.0	51.0	50.0	49.8	50
BoolQ	56.0	47.8	54.3	46.5	63.0	50
OpenBookQA	13.5	13.0	16.0	12.5	21.0	25
SciQ	66.5	69.5	48.5	66.5	84.3	25
COPA	51.0	53.0	56.0	56.0	67.0	50

Tool-only SFT (D) harms SciQ (−18 pp vs C) while helping some two-way tasks (Figure 7). Multi-mix SFT restores SciQ and keeps macro near C: CE replay is required when adding tools at this scale. Further dense continue (C+) specializes: SciQ +3.0 and ARC-Easy +2.5, BoolQ −8.2, macro flat. Val CE never beat the C resume bar on C+, yet end-of-run weights improved SciQ (multi-domain best-val is a false friend).

7.3 Agent battery

Table 6 uses tool-SFT ship weights (pure_v2). Read pure first: freeze pure is 90/101 with no_call 15/24 (stream-visible scoring on the ship pin). Product 101/101 is a system score (assist off, host gate on).

Gate decomposition (instrumented re-run). On the same pure_v2 pack with host-gate event logging (battery_*_gatestats.json):

- Agent protocol cases: 85 (of 101 total with host/lang/tok checks).

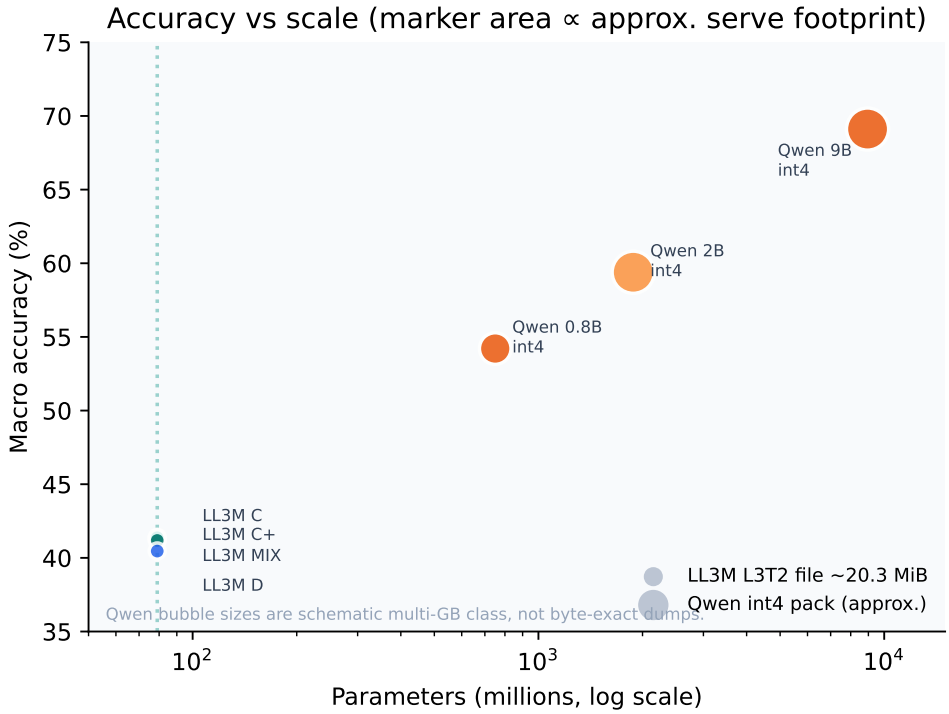


Figure 6: Accuracy versus parameter scale. Marker area is proportional to approximate serve footprint (LL3M L3T2 file 20.3 MiB; Qwen bubbles schematic multi-GB class).

Table 6: Closed 101-case agent battery (C L3T2, greedy). Product = assist off, host gate on.

Lineage	Policy	Total	calc	lookup	no_call
pure_v2	product	101/101	30/30	20/20	24/24
pure_v2	pure	90/101	30/30	20/20	15/24
denser C	pure	42/101	0/30	0/20	24/24
D	pure	37/101	3/30	7/20	8/24
MIX	pure	70/101	24/30	19/20	2/24

- Model emitted a parseable <call> on 85/85 protocol cases.
- Gold host intent needs a tool on 58 cases; the model called on 58/58 (0 silent).
- On 27 no-tool-intent cases the model still called; with gate on, the host blocked all 27 (product no_call stays 24/24).
- Host rewrote a wrong tool name to gold intent 2 times.
- With gate off, the same 27 false-call emissions are not blocked: emission-aware pure is 72/101; final-text-only recovery is 99/101.

Thus product no-call is primarily a host gate result (§7.4); the freeze pure pin (90/101) sits between emission-aware and final-text-only pure. A prior same-flag product re-run scored 99/101 (2/5 time); pins under paper_freeze_v072/. Suite schema: docs/AGENT_BATTERY_SUITE.md. Train–serve greedy match on the tool ship pack: 16/16 tokens (C golden).

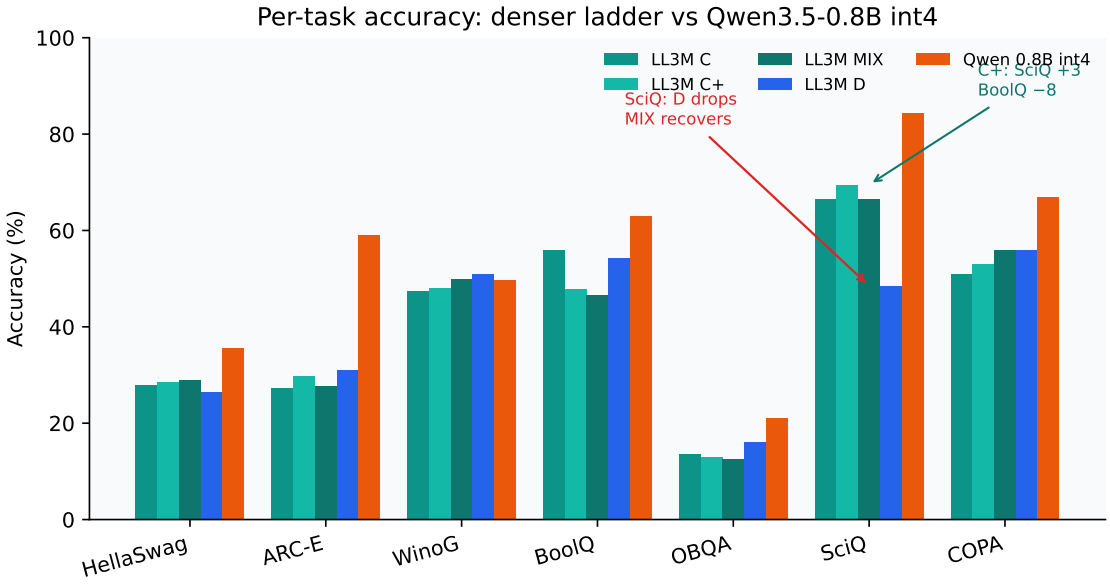


Figure 7: Per-task accuracy including C+. Multi-mix recovers SciQ after tool-only SFT regression while staying near denser-continue macro.

7.4 Policy–knowledge Pareto

Short tool-heavy QAT hits a false-call basin (Table 7). Denominators shown; SciQ for d12/s2 from SciQ-only runs (same seed/limit).

Table 7: Gate-off pure policy vs SciQ under short QAT (denominators explicit).

Lineage	calc	lookup	SciQ	no_call	Note
denser C	0/30	0/20	66.5	24/24	no tools
pure_v2	30/30	20/20	57.75	15/24	ship
sciq_trade	14/30	14/20	67.25	9/24	SciQ twin
C+	—	—	69.5	—	SciQ peak
d12 flood	30/30	20/20	60.3	7/24	overtool
s2	27/30	19/20	65.8	8/24	SciQ→tools
MIX	24/30	19/20	66.5	2/24	CE replay

Implication. Dual ship (agent pack + SciQ/weight twin) and/or product gate are design outcomes under the CCD budget, not temporary engineering debt. Recovering pure no-call after tool-mass collapse is hard with CE alone.

Preference follow-up (P3). Naive DPO on false_call pairs (1k steps) reaches pure no_call 24/24 but destroys tools and fluency. Careful DPO (150 steps, ∼55% tool CE retain, low β) raises pure total 90→93/101 (no_call 23/24) while product gate-on remains strongest on pure_v2 (101/101). Product no-call is therefore primarily a host gate result; pure no-call is a research ceiling near the mid-90s without a better multi-objective preference recipe.

7.5 Residency

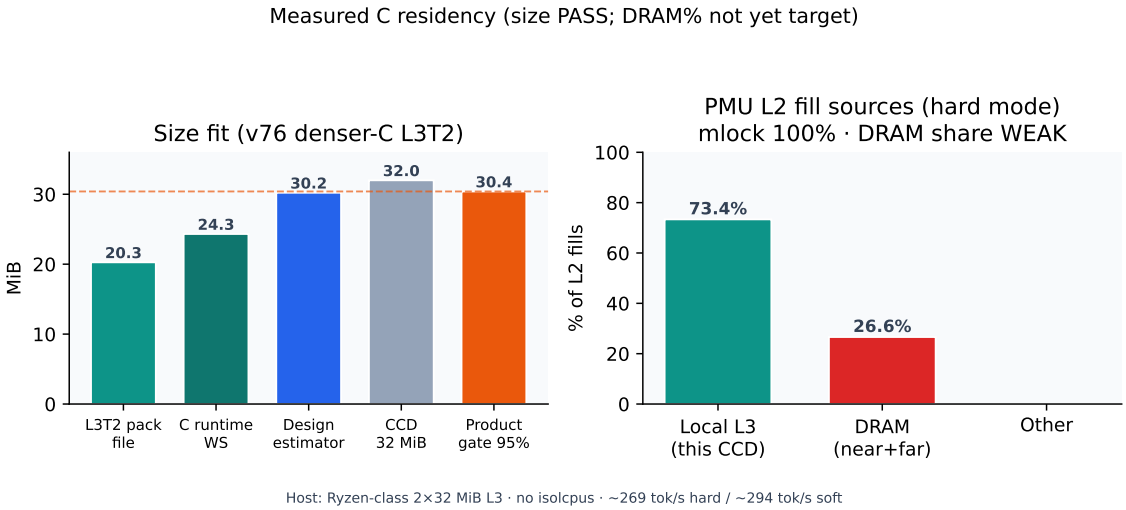


Figure 8: Measured C residency on denser-C L3T2: size-fit PASS (runtime WS 24.3 MiB < 32 MiB CCD), mlock 100%, hard+FIFO can reach GOOD DRAM% (≤5%) on long windows; soft multi-core is not faster.

Table 8: On-host C decode measurement (denser-C pack, reference Ryzen 2×32 MiB L3).

Metric	Soft	Hard
C runtime WS (MiB)	24.32	24.32
L3T2 weights payload (MiB)	20.27	20.27
L3T2 on-disk file (MiB)	20.32	20.32
Fits 32 MiB CCD	yes	yes
mincore mlock	100%	100%
tok/s (preferred)	∼260–270	∼280–282
Best hard L3 share of L2 fills	—	up to ∼97%
DRAM share of L2 fills	∼30–40% often	≤5% GOOD / WEAK
SCHED_FIFO	no	OK (cap_sys_nice)

Table 8 and Figure 8: size-fit and mlock hold; hard+FIFO is preferred once cap_sys_nice is available. Boot isolcpus remains the path to more reliable near-zero DRAM.

Multi-session under the same gate. Shared weights and independent states obey $WS = W + N \cdot S(\text{kv_cap})$. Full T=256 admits N=2 (∼550 tok/s aggregate); N=3 is refused. With shorter KV, N=8 at kv_cap=64–128 reaches ∼2100–2200 aggregate; N=16 short-KV peaks ∼2300 aggregate (Table 9).

Table 9: Multi-session packing under the 30.4 MiB CCD gate (shared L3T2 weights).

Config	WS (MiB)	tok/s / stream	Aggregate
N=1, kv 256	24.3	∼280	∼280
N=2, kv 256 (max full T)	28.4	∼274	∼550
N=3, kv 256	32.4	—	refused
N=8, kv 128	29.6	∼270	∼2100
N=8, kv 64 (best balanced)	25.2	∼277	∼2190
N=16, kv 32–48 (peak agg.)	∼26–28	∼146	∼2300

8 Discussion

The −12.8 pp closed-book gap vs Qwen3.5-0.8B portable weight-only int4 is largely the price of the CCD gate (plus less data and a tiny vocab). The relevant metric is quality per CCD-resident MiB and agent reliability under host tools, not iso-param MMLU.

Tool-only SFT under a tiny ternary student overwrites educational surface statistics (SciQ −18 pp; pure agent 37/101). Multi-mix SFT restoring SciQ is systems evidence that policy learning must share bandwidth with dense CE when capacity is CCD-bounded. Figure 4 is the knowledge-placement decision procedure when the gate is full.

We separate six budgets: train scale, pack file size, infer WS (design vs runtime), PMU DRAM share, agent policy flags, and multi-session N. CCD-gated co-design forces that distinction. Future work: isolcpus-grade DRAM%, same-shape fp32 MC, bitnet.cpp/llama.cpp latency peers, full multi-seed macro grids (SciQ limit-100 probe only so far), and multi-objective preference learning under QAT. KB on/off for lookup is measured (20/20→0/20).

9 Limitations

Closed-book MC is not the product objective; T=256 and V=2048 limit long documents; LL ranking vs Qwen is not iso-tokenizer. We did not run multi-trillion-token overtraining or full teacher KD. Qwen comparisons are iso-task under weight-only int4, not iso-data; NF4 and public ∼100M SLMs are absent. Residency with isolcpus is host-dependent; without it DRAM% is often WEAK even when size-fit passes. The agent suite is closed (101 cases), not BFCL; product scores include the host gate. No pure-policy pack jointly maximizes tools, SciQ, and pure no-call under short QAT. Primary MC is single-seed limit 400; a SciQ limit-100 three-seed probe has stdev ∼1.2 pp. Incomplete GPU-hour ledger; design vs runtime WS differ (both fit the gate).

10 Conclusion

We named and operationalized CCD-gated co-design: treat one CPU L3 CCD as the product unit and jointly choose shape, ternary/int8 QAT, C packing, dense data, and host Engram-lite knowledge so the full decode WS size-fits. LL3M is a concrete operating point (∼79M, 20.3 MiB pack, design WS ∼30.2 MiB, runtime WS ∼24.3 MiB) with continuous train–serve lock. Closed-book macro trails Qwen3.5-0.8B portable weight-only int4 by −12.8 pp absolute on this suite while remaining CCD-gated; multi-mix SFT preserves science signal that tool-only SFT destroys. On the tool ship pack, report pure 90/101 and product 101/101 (host gate on) as dual metrics on a closed suite—not BFCL. Size-fit and mlock are measured on the reference host; hard+FIFO holds ∼280–282 tok/s; DRAM% is often WEAK without isolation. Multi-session packing reaches ∼2k aggregate tok/s at N=8 short KV. The path forward is host KB growth, gate-intervention instrumentation, multi-mix or preference learning, multi-session serving, and isolcpus-grade residency—not encyclopedias in ternary weights.

Ethics statement

A ∼20 MiB pack enables offline CPU use (privacy, edge) and unsupervised local generation. Closed tools and host gates limit open-ended tool abuse; data licenses and a model card will accompany any public release.

Reproducibility

Artifacts are listed in Appendix B. Train used remote CUDA; serve claims target portable C on the L3 reference host. Reproduce agent scores from checkpoints/paper_freeze_v072/ (SHA256 in MANIFEST.json).

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A Hyperparameters

Dense continue (C). Resume ternary base step 60k; absolute steps 90k; Lion; lr 10^{-4}, warmup 500; microbatch 8× accum 8 (16 384 tok/step); dropout 0.1; CUDA train.

Multi-mix SFT. From C; +5k steps; lr 3×10^{-5}; mix ≈50/20/30 dense/fact/tool by character budget; QAT ternary+int8.

B Artifact paths

Artifact	Path
Shape freeze	checkpoints/v75_freeze.json
Paper freeze (KB + agent)	checkpoints/paper_freeze_v072/ (see MANIFEST.json SHA256)
Ladder / residency docs	docs/DENSE_*, docs/L3_RESIDENCY.md
MC JSON (C / C+ / D / MIX)	checkpoints/v76_dense_*
Qwen int4 aggregate	checkpoints/compare/ll3m_vs_qwen35_family_int4.json
Residency + multi-session	checkpoints/compare/
Multi-stream binary	c_infer/multi_bench
QAT vs post-hoc	checkpoints/v75_ternary/quality_compare.json
C runtime	c_infer/
Markdown working draft	docs/paper/LL3M_L3_Resident_Ternary_SLM.md

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